

به نام خدا



## معماری کامپیوتر

نیمسال دوم ۰۳-۰۴

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فروردین ۱۴۰۴

مبانی معماری  
کامپیوتر

جلسه حل تمرین

### سوال یک

T0:  $AR \leftarrow PC$   
T1:  $IR \leftarrow M[AR], PC \leftarrow PC + 1$   
T2: decode, Indirect  $\leftarrow I$   
011  $\rightarrow$  STA :  $D3 \leftarrow 1$   
T3:  $AR \leftarrow M[128]$   
T4:  $M[AR] \leftarrow AC, SC \leftarrow 0$

### سوال دو

PUSH A  
PUSH B  
ADD  
PUSH C  
MUL  
POP D

### سوال سه

LOAD R1, x  
LOAD R2, y  
ADD R3, R1, R2  
STORE z, R3

Answer: 3

### سوال چهار

$DR \leftarrow M[512]$  // Load  
 $DR \leftarrow AC, AC \leftarrow DR$  // swap(AC, DR)  
 $AC \leftarrow \neg AC$   
 $AC \leftarrow AC + 1$  // Two's complement  
 $AC \leftarrow AC + DR, SC \leftarrow 0$

## سوال پنج

$$\text{Address} = \text{base} + \text{offset} + \text{index} \times \text{scale}$$

- $100(R2)$ : offset 100 from the value in  $R2$
- $R1 = R1 + \text{Memory}[100 + R2 + d \times R3]$
- Given:  $R1 = 5, R2 = 200, R3 = 2, d = 4$
- Calculate:

$$100 + R2 + (d \times R3) = 100 + 200 + (4 \times 2) = 308$$

## سوال شش

| Instruction Type | Operand Types Used     | Register Bits Used |
|------------------|------------------------|--------------------|
| 16-bit           | 3 registers            | $3n_{regb}$        |
| 32-bit           | 2 registers + 1 memory | $2n_{regb}$        |

$$n_{regb} = \log_2(\text{number of registers})$$

For 16-bit format:

$$op_1 + 3n_{regb} = 16$$

For 32-bit format with memory addressing (memory has  $2^{20}$  addressable units):

$$op_2 + 2n_{regb} + 20 = 32 \implies op_2 + 2n_{regb} = 12$$

Since the number of instructions in both formats is the same,  $op_1 = op_2$ , so:

$$16 - 3n_{regb} = 12 - 2n_{regb} \implies 4 = n_{regb}$$

Therefore:

$$n_{regb} = \log_2(\text{number of registers}) = 4 \implies \text{number of registers} = 2^4 = 16$$

## سوال هفت

### Instruction Fetch Phase:

1. PC  $\rightarrow$  AR: AR  $\leftarrow 3AF$
2. Memory Read: IR  $\leftarrow M[AR] = 932E$
3. Increment PC: PC  $\leftarrow 3B0$

### Instruction Decode Phase:

| Bits  | Field            | Meaning                     |
|-------|------------------|-----------------------------|
| 15    | I (Indirect bit) | 0 = direct, 1 = indirect    |
| 12-14 | Opcode           | Operation code              |
| 0-11  | Address          | Address part of instruction |

IR = 932E (hex) → binary: 1001 0011 0010 1110

- I = 1 (indirect)
- Opcode = 001 (ADD)
- Address = 32E

Instruction: ADD I 32E

### Effective Address (Indirect):

- Memory Read:  $M[32E] = 09AC$
- $AR \leftarrow 09AC$

### Execution Phase: ADD

- Memory Read:  $M[09AC] = 8B9F$ , so  $DR \leftarrow 8B9F$
- Add:  $AC = 7EC3$ ,  $DR = 8B9F$
- $7EC3 + 8B9F = 10A62$  (hex)
- Result in AC: 0A62 (lower 16 bits)
- Carry out (Extend bit E) = 1

### Final Register Values:

| Register | Final Value | Explanation                                    |
|----------|-------------|------------------------------------------------|
| AC       | 0A62        | Result of addition with carry out              |
| DR       | 8B9F        | Value read from effective address              |
| AR       | 09AC        | Final effective address                        |
| IR       | 932E        | Instruction fetched                            |
| PC       | 3B0         | PC incremented after fetch                     |
| I        | 1           | Indirect addressing bit                        |
| E        | 1           | Carry out from addition                        |
| SC       | 0           | Step counter reset, ready for next instruction |

| Opcode | Mnemonic | Operation                                                        |
|--------|----------|------------------------------------------------------------------|
| 000    | AND      | $AC \leftarrow AC \wedge M[addr]$                                |
| 001    | ADD      | $AC \leftarrow AC + M[addr]$                                     |
| 010    | LDA      | $AC \leftarrow M[addr]$                                          |
| 011    | STA      | $M[addr] \leftarrow AC$                                          |
| 100    | BUN      | $PC \leftarrow addr$                                             |
| 101    | BSA      | $M[addr] \leftarrow PC; PC \leftarrow addr + 1$                  |
| 110    | ISZ      | $M[addr] \leftarrow M[addr] + 1$ ; if zero skip next instruction |

**Additional Notes:**

- Address 0001 (decimal 1) is often used to store the address of the interrupt service routine.
- FGO = 1: Output device ready to accept data from CPU.
- FGO = 0: Output device busy; CPU must wait.
- IEN = 1: Interrupts enabled; CPU can respond to I/O interrupts.
- BUN 2300: Branch unconditionally to address 2300.
- ION: Instruction to enable interrupt handling from I/O devices.
- BUN 0 I: Branch unconditionally to address stored at memory location 0 (indirect mode).